

Question 2: Explain the purpose of the new keyword in Java.

The new keyword in Java creates a new instance of an object, giving it a memory allocation. Every time you use this keyword, a new object is created, which can then be used to perform various operations, store data, or call methods.

Question 3: Explain the difference between a syntax error, a logic error, and a runtime error. Then, write a short fragment of Java code to demonstrate each type.

Syntax errors occur when one of Java's fundamental grammar rules are broken, resulting in being caught by the compiler and the build will fail to load.

```
public class SyntaxError {  
    public static void main(String[] args) {  
        int number = 10  
        System.out.println(number);  
    }  
}
```

Runtime errors only occur after the code has been successfully built, however the Java Virtual Machine is faced with an impossible operation, resulting in an ArithmeticException error and the program crashes.

```
public class RuntimeError{  
    public static void main(String[] args) {  
        int a = 5;  
        int b = 0;  
        int result = a / b;  
        System.out.println(result);  
    }  
}
```

Logic errors only occur when the programmer messes up with displaying outputs, and the only problem with that is you may have wrong or inaccurate data displayed on your program. This doesn't affect the success of your program's builds, nor result in any crashes throughout the project.

```
public class LogicError{  
    public static void main(String[] args) {  
        System.out.println("Hello Wrld!");  
    }  
}
```